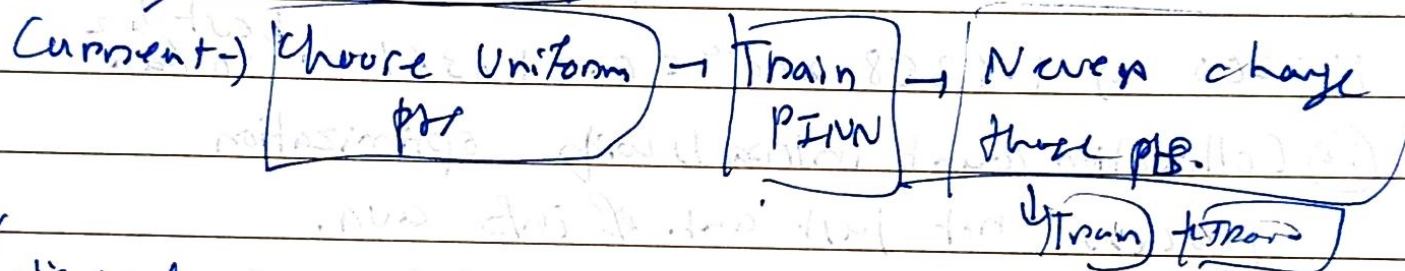


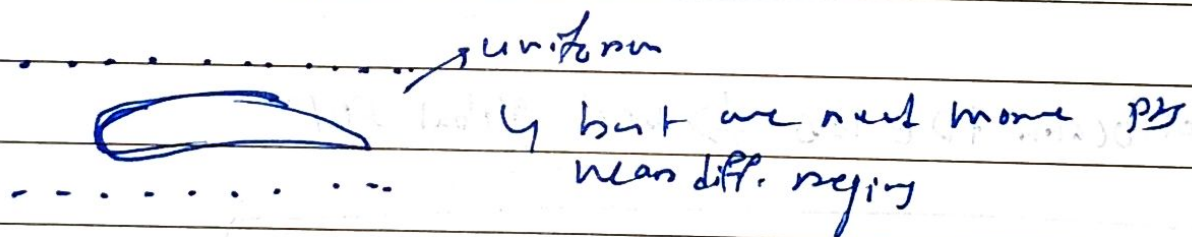
tommo → Read sampling papers fix

(Wu, papers) (sampling)
① Comprehensive Study of non-adaptive ...
~~fixed residual pts~~



↳ "Satisfy PDE at these particular locations"

↳ But it may be doing poorer job b/w these locations



② RAR (Residual Based Adaptive Refinement (RAR))

↳ How many pts. should I guess?

③ to
"where is my current PINN failing for
satisfying the physics"

$x_{colloc} = \text{touch. lnspace}(\dots)$

↳ every epoch we have same ^{Date: pths.}

BASIC idea :-

Start with $\rightarrow$ Train $\rightarrow$ check PDF $\rightarrow$ find loc. $\rightarrow$ Add those
Sparse pts. PZIVIV residual over many with loc. with as
candidate location large residuals large new
colloc. pts.

$\rightarrow$ Train $\rightarrow$ Repeat
again

① Train $\rightarrow$ Find error $\rightarrow$ Add $\rightarrow$ Retrain
pts

② domain $\Rightarrow 0 \leq x \leq 1$

$N = 20$

After training $\Rightarrow$

$x = 0.1$ residual = 0.001

$x = 0.3$ residual = 0.002

$x = 0.55$ residual = 0.80 $\leftarrow$ BAD

$x = 0.8$ residual = 0.003

RAN says $\rightarrow$ network is struggling among $x = 0.55$ put
more many pts there:

③

.....

.....

↳ high error
area

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⑥ follow ptr $\Rightarrow R(n_i)$

For $\rho(x) = 1$: $R(x) = U_{\text{even}}(x) + \pi^2 \sin(\pi x)$

$$L_{PDE} = \frac{1}{N} \sum_{i=1}^N R(m_i)^2$$

↳ more ppt res. smth at the selected loc.

↳ Problem $\Rightarrow$ PDE exist over whole conti. domain not only at those N pts.

④ loc. & distribution of those pts. are highly imp. to performance

→ maybe per imp. ques. isn't only how many
pts. we use but where we put them.

⇒ RAR → Lu et al. 2019 → this paper refers it to as RAR-G → greedy selection behavior.

★ Train f_{test} . And where PDE residual is large, put more training pts. there and train again

Start $\rightarrow$ Choose initial $\rightarrow$ Train $\rightarrow$ Search $\rightarrow$ calculate PDE

space	colloc.	PINN	many candidate	res. at candidate
PF			locating	

→ Find loc. with largest residual → Add term to train. set → Train again → Repeat.

Let $N=20 \rightarrow$

n	residual
-0.5	0.002
-0.4	0.001
0.0	0.003
0.3	0.005
0.7	0.8 ← huge

50 => $\dots$
 $\dots$
 $\dots$ $\xrightarrow{\text{now}}$ $\dots$
 $\hookrightarrow$ diff. neg

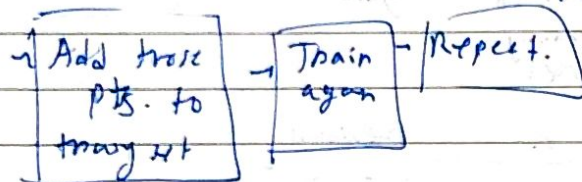
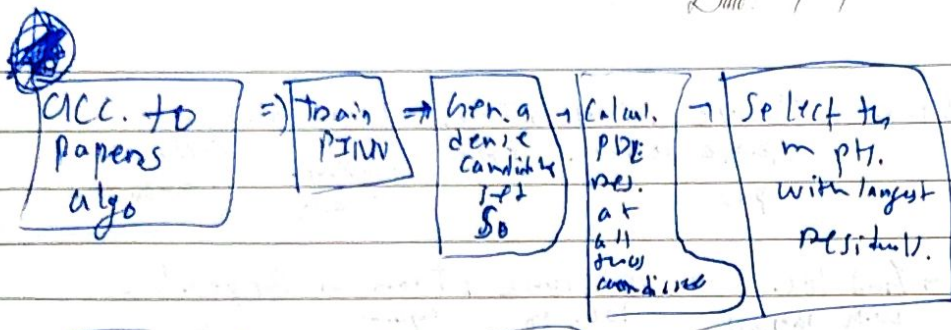
(*) How RAR knows where the network is struggling?

↳ you can't eval. the PDE at every pt. in conti. domain

↳ So $align_first_created =$

Candidate set = 1 So

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(i) existing training pts + temp. candidate pts

- greedy. deciding where to put new $\Rightarrow |R(n)|$

why greedy $\Rightarrow$ coz it focuses heavily on largest residual loc.

Let 0.1, 0.2, 0.3, 0.9, 0.95

RAR-G might $\Rightarrow$ 0.95, 0.9 (basic RAR)

RAD & RAR-D $\rightarrow$ proposed by authors

RAD provides higher sampling probability to regions where the current PDE residual is high.

RAD $\rightarrow$ Residual Based Adaptive Distribution:

"Give higher prob. to locations with larger residuals, but don't completely ignore the rest"

$$p(n) \propto \frac{E^k(n)}{E[E^k(n)] + C}$$

Probability density

$k \uparrow \Rightarrow$ more pts around high resid. regions

avg value of E^k over the domain

$E(n) \Rightarrow$ mag. of residual
 k = controls how strongly we favor high-res. regions.

C = Controls how much we retain a tendency toward uniform sampling.

$C \uparrow \Rightarrow$ controls how much distribution more uniform



distri. sam. which reg. has more resi. than other.

Uni $\rightarrow$

RAD $\rightarrow$
 $\uparrow$ more probability

$k=0 \rightarrow$ goes uniform behavior

(i) pts clusters statistically around diff. regions rather than only selecting single worst areas.

$k \uparrow C \downarrow \Rightarrow$ more spread throughout the domain

$k \downarrow C \uparrow \Rightarrow$ more spread throughout the domain

Anderson
 $k=1, C=1$
 per. useful

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① RAR-D → combine both ideas

↳ Residual based Adaptive Refinement with distribution.

RAR → keep adding pts

RAD → select new pts. acc. to a residual based prob. distribution.

Algo $S \leftarrow$ Sample pts. acc. to $p(x)$

$T \leftarrow T \cup S$

② → add the newly selected pts. to existing training set.

③ RAR → add works... hierarchy...
RAD → resample acc. to residual probability
RAR-D → add pts. acc. to residual probability

④ Which one did perform best?

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- RAD perf. overall among the term sampling methods across the form. & inn. process:

→ RAD & RAR-D → more eff. for complicated soln

↳ ② won't be much eff. for our smooth case.

④ goal is to learn mech. for harder PDE's

⑤ Testy on Burgers' eqn

$$U_t + U U_x = \nu U_{xx}$$

Sharp front near $x=0$ at $t \approx 1$.

at 2000 res. ⇒ result was.

pt1 L2 error

Random = 13.3%

Hammersley = 3.02%

③ RAR-G = 0.12%

② RAD = 0.02%

① RAR-D = 0.03%

→ the more we use adaptive methods can conc. res. pts eff.

① This paper imp. recognizes PINN have randomness
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RAR/RAD says

↳ ~~the~~ comp. PDE res. tell us where the diff. region is:

② Our ex: $N = 10, 20, 50, 100, 200 \rightarrow$ change in qty

RAR $\rightarrow$ change in distribution. ✓

③ RAR doesn't mean more pts. ☹

Let $\rightarrow N = 100$ pts

RAR might start time and add:

20 $\rightarrow$ high residual pts.

now $\rightarrow 120$

then $+ 20 \rightarrow 140$

then $\rightarrow 160$

④ RAR can inc. pts over time but then new pts. inc. bias:)

the new pts are chosen intelligently

(RAR-h)

Random-R

① Uniform sampling with resampling:

instead of $\rightarrow$ pts A $\rightarrow$ train $\rightarrow$ pts A $\rightarrow$ train $\rightarrow$ pts A

↳ periodically gen. new random pts

$\rightarrow$ pts A $\rightarrow$ train $\rightarrow$ pts B $\rightarrow$ train $\rightarrow$ pts C $\rightarrow$ train

↳ Random R

② Now $\rightarrow$ Vanilla PINN $\rightarrow$ fixed uniform grid:

super $\rightarrow$ Grid $\rightarrow$ Halton

$\rightarrow$ Random $\rightarrow$ Hammersley

$\rightarrow$ LHS $\rightarrow$ Sobol

$\rightarrow$ Random-R

$\rightarrow$ RAR-h, RAD, RAR-D

③ Auto strategy

$\rightarrow$ PINN results have randomness from:

↳ Random sampling, network initialization, optimization.

$\therefore$ try run each exp atleast 10 times & calc. geo. mean & std. dev.

our exp $\rightarrow$ learning experiments

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Understanding from papers & exp

→ imp. problem is not simply the no. of colloc. pts; it is whether the residual pts are distributed effectively over the domain.

① next exp =) Vanilla VS RAR-5 ^(page 6)
Fixed Uniform

fixed pts
↓
train
↓
results

VS

ini. pts
↓
train
↓
find high res. areas
↓
add pts there

Compare =) L_2 , L_{PDE} , LBC , no. of pts., training time

RAR-5 algo in paper

(S1) $T \rightarrow$ ini. set of residual pts.

(S2) Train pinns for some iterations.

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(S3) Construct dense candidate set : $S_0 \rightarrow$ search set

(S4) Calculate PDE at every point in S_0 .

(S5) select m candidates having the largest residual.

(S6) Add them to existing training set: $T \leftarrow T \cup S$
select new training pts.

(S7) Train again.

(S8) Repeat.

Basin 1. 10 pts $\rightarrow$ train pinns $\rightarrow$ Candidate set (1000 pts)
 $\rightarrow$ calc. res at all 1000 $\rightarrow$ pick worst 5 $\rightarrow$ add these 5 to training set.

We use these 1000 pts to know where pinns is struggling

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★ RAR-G tradeoff.

↳ it attacks highest resi. pts., it & C
but C is maybe also imp. but
not the largest.

↳ ∴ RAD → which uses residual-based prob.
distribution instead of only selecting
largest residual loc.

extra sampling / search cost. ↔ better acc. with
fewer inte. places
pts.

★ FRA: residual based prob.

↳ def. how likely
diff. locations
are to be
selected.
 $E(n) = |R(m)|$
↳ PDE model

Distribution → Diff. parts of domain get
diff. probs

PDE → sampling prob. → when new
ptr. go

② imp. just did we ask, which supply
could be so much - on volume Date: / /

Analysis



20 diff. locs
for police
in a city

→ patrol the city

↳ but where trouble is highest → patrol 1st - an
Cand. set.

↳ so if there are 5 areas have the most
trouble, assign off. there.

↳ 5 → new way to

★ is so random or uniform?

↳ can be either

↳ depends on ~~each~~ method choice.

↳ equipm, names (MS, Malton, Marnesley,
Solihull).

Results

↳ RAD perf. better among 10 sampling
methods for all

↳ forward → diff. bungal, Allen-Cahn,
wave.

↳ inverse → diff. rxn, Kosteng de Vries
when in 6 boundary probl. & for exp.
Setup, RAD had the best perf.

↳ RAD → strongest overall results!

Ⓐ The usefulness of adaptive sample depends strongly on the diff. & structure of solⁿ

exp $\Rightarrow N=20 \Rightarrow L_2 = 8.5618 \times 10^{-4}$
fixed uniform

$N=20$ (RAR.4) $\Rightarrow L_2 = 3.1558 \times 10^{-4}$

63% $\downarrow$ error

but $\rightarrow$ Initial pts(1) = 20

$20 \rightarrow 25 \rightarrow 30 \rightarrow 35 \rightarrow 40 \rightarrow 45$

5 ~~pts~~ ^{worst} at every epoch _{from 2000} So

$\therefore T \Rightarrow TUS \Rightarrow \underline{N=45}$

Experiment

Convergence $\rightarrow$

why prime? $L_{PR} = \frac{1}{12} \sum_{i=1}^{20} R(n_i)^2$

after $L_{PR} = \frac{1}{25} \sum_{i=1}^{25} R(n_i)^2$

Ⓐ optimizes now seeing diff. loss function

Experiment

ⒸA

$N=100$

Fixed uniform

$L_2 = 1.143 \times 10^{-3}$

$L_2 = 1.143 \times 10^{-3}$

ⒸC

RAR starting from

100

$N_{ini} = 100$

$100 \rightarrow 105 \rightarrow 110 \rightarrow 115 \rightarrow 120 \rightarrow 125$

$\hookrightarrow 100 \rightarrow 125$ RAR

change / add

$\Rightarrow N_{target} = 125$

so it doesn't rely on epoch numbers. And actually stop at 125

ⒸB

$N=125$

Fixed & Uniform

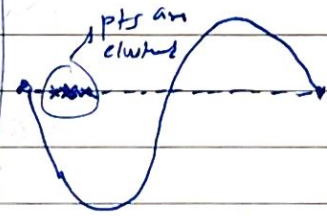
$L_2 = 1.6957 \times 10^{-4}$

added $\Rightarrow$ $IN_target = 125$ $IV = 10112$
100pts RAR-LS
col

epoch $\rightarrow$ 500 $\rightarrow +5 \rightarrow 105$
 700 $\rightarrow +5 \rightarrow$
 1500 $\rightarrow +5 \rightarrow$
 2000 $\rightarrow +5 \rightarrow$
 2500 $\rightarrow +5 \rightarrow 125$
 3000 $\rightarrow +5 \rightarrow 125$

why if epoch $\neq$ par-period
 $= 0$ & $x_colloc_sphere[0]$
 $+ m_add \neq IN_target$

(i) at 3000 $\Rightarrow 125 + 5 \neq 125$
 failure: only 125 no further refinement

Uniform $\Rightarrow IN = 100$	Uniform $\Rightarrow IN = 125$	$N = 100 \rightarrow N = 125$ adaptive
$L_2 \approx 1.143 \times 10^{-3}$	$L_2 = 1.699 \times 10^{-4}$	$L_2 = 3.839 \times 10^{-5}$
		pts are clustered  $x \rightarrow 2$ adaptive pts.

Conclusion $\Rightarrow$

$L_2(C) \approx 3.34 \times 10^{-3}$ & $L_2(B) \approx 1.70 \times 10^{-4}$

(i) 19.6 x larger than Case B
 Case C $\Rightarrow$ 2.9x worse than Case A

Why RAR-LS failed

Greedy $\rightarrow$ 25 pts are clustered not distributed across the domain

also $\rightarrow$ adding 25 pts recovered the accuracy than adding 25 pts adaptively

Under specific setup 1D non setup $\rightarrow$ RAR-LS performed worse.

papers itself presented RAD/RAR-D takes to get more control over distribution pts.

Ⓐ Burger's Eqⁿ (understanding)

↳ Poisson's → simple

↳ Burger's → time dependence
↳ Non-linearity

Ⓐ From sampling papers (standard benchmarks)

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

time $\frac{\partial u}{\partial t}$ $\frac{\partial u}{\partial x}$ non linear term

with $x \in [-1, 1]$, $t \in [0, 1]$

I.C $\Rightarrow u(x, 0) = -\sin(\pi x)$

B.C $\Rightarrow u(-1, t) = u(1, t) = 0$

& $\nu = \frac{0.01}{\pi}$

this develops a very steep region near $x=0$ as t approaches 1

Ⓐ $a(x) u_x$
already known $\frac{\partial u}{\partial x}$
↳ (ie) linear in u

$\Rightarrow \frac{\partial u}{\partial t} \Rightarrow u$ changes as time passes

solⁿ $\Rightarrow u = u(x, t)$

problem $\Rightarrow u = u(x)$

N.N learners $\Rightarrow (x, t) \rightarrow u$

inputs of $\Rightarrow x \rightarrow u$

Non-linear term $\Rightarrow u \frac{\partial u}{\partial x}$

u multiply its own derivative this makes PDE non-linear.

Poisson's $\Rightarrow u_{xx} = -x^2 \sin(\pi x)$

↳ In PDE residual depends on derivatives of NN output but no product like:

$u_0 u_1, x_0$

$R(x, t) = u_t + u u_x - \nu u_{xx}$

↳ in burger's the network's prediction itself enters the residual

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① diffusion & viscosity term

$v_{u_{xx}}$ (RHS)

$v \left(\frac{\partial^2 u_0}{\partial x^2} \right)$ → this is what needs to be calculated which is similar to Poisson using auto diff

(ii) $\left(\frac{\partial u_0}{\partial t} \right)$ → ~~$\frac{\partial u_0}{\partial t}$~~ is the new derivative we need

$v \rightarrow$ viscosity / diffusion coeff.

$$v = \frac{0.01}{n}$$

u_{lin} vs v_{lin}
non linear term distort the profile while v smooths it

at small v soln develops sharp region

Poisson $\Rightarrow n \rightarrow PINN \rightarrow u$

Burger $\Rightarrow \begin{bmatrix} n \\ t \end{bmatrix} \rightarrow PINN \rightarrow u$

So $\rightarrow$ nn: Linear (1, 32)

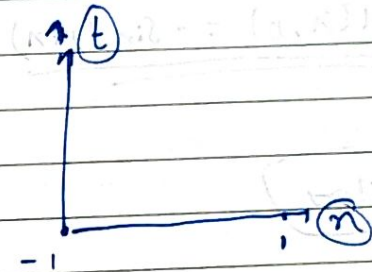
nn: Linear (2, 32)

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12 i/p $\rightarrow$ 32 $\rightarrow$ 32 $\rightarrow$ 32 $\rightarrow$ 1

Domain $\Rightarrow$ Poisson
 n
-1

Burger $\Rightarrow$



$(n, t) \in \underline{[-1, 1]} \times \underline{[0, 1]}$ $\rightarrow -1 \leq n \leq 1$
 $\rightarrow 0 \leq t \leq 1$
means all possible combi. of n & t

Colloc. pts $\Rightarrow n_1, n_2, n_3, \dots$

$\hookrightarrow (n_1, t_1), (n_2, t_2), (n_3, t_3), \dots$

t as second dimension

Initial conditions

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Poissons: $\underline{u(-1) = 0}$ & $\underline{u(1) = 0}$

Burger $\rightarrow$ at $\boxed{t=0}$

$\underline{u(x, 0) = -\sin(\pi x)}$

Boundary conditions

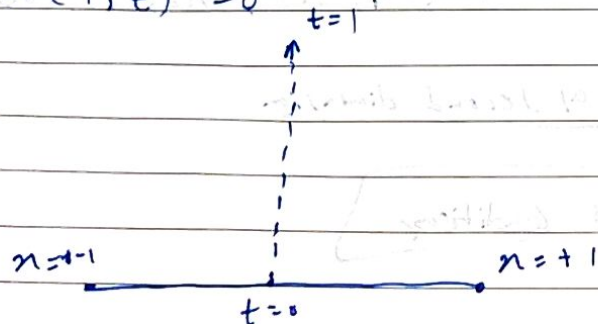
at $\underline{x = -1}$ & $\underline{x = 1}$

for all relevant times:

$u(-1, t) = 0$

& $u(1, t) = 0$

Initial



$u(x, 0) = -\sin(\pi x)$

MATRIKAS

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3 diff.

training location

PDE/collocation pts

Ini. condi. pts

B. Cond. pts.

Loss

Poisson: $L = L_{PDE} + L_{BC}$

Burger: $L = L_{PDE} + L_{IC} + L_{BC}$

we use

PDE loss $\Rightarrow$ At internal (x, t) pts:

$L_{PDE} = \text{MSE}(u_t + uu_{xx} - v u_{xx})$

IC loss $\Rightarrow$ $\hat{t} = 0 \Rightarrow L_{IC} = \text{MSE}(u_0(x, 0) + \sin(\pi x))$

$\therefore -\sin(\pi x)$

gen. error

B.C loss $\Rightarrow L_{BC} = \text{MSE}(u_0(-1, t), 0) + \text{MSE}(u_0(1, t), 0)$

adjust weight

Loss on equ. combination formulation
just complicated
 $\Rightarrow$ prediction $\rightarrow$ derivatives $\rightarrow$ residual $\rightarrow$ loss $\rightarrow$ backpropagation

RAR, RAD, RAR-D EXE

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Burgers & RAR

Poisson $\Rightarrow$ Smooth $\Rightarrow u = \sin(\ln u)$

$\hookrightarrow 100, 125, 150 \rightarrow 125$
 $\hookrightarrow$ worst case

diff. situation for burgers

$\hookrightarrow$ Burgers dev. steep region & in the study adaptive RAD & RAR-D had better acc. than uniform for diff. soln

$\hookrightarrow$ Burgers $\rightarrow$ 2000 residual pts. For sampling comparison & found that uniform approaches struggled to capture the soln well, where adaptive methods achieved much lower L_2 error.

"Maybe adaptive sampling didn't help pass because it was too smooth & maybe it will help for difficult soln"

test

$\Rightarrow$ Underlying $\rightarrow$ Simul pin $\rightarrow$ Vents $\rightarrow$ obs time $\rightarrow$ conversion.
Failure $\rightarrow$ Lit $\rightarrow$ imp $\rightarrow$ Comp. $\rightarrow$ progress Date: / /

Analysis $\rightarrow$ this banner $\rightarrow$ no. of pts
has its own $\rightarrow$ optimizers
 $\hookrightarrow$ Adam/L-BFGS solver
 $\hookrightarrow$ sampling methods
 $\hookrightarrow$ coarsening permits
 $\hookrightarrow$ evaluation procedure

② I will build my own progression.

$\hookrightarrow$ ① Make Vahilla Burgers PENIN work

$\Rightarrow$ SB1 $\Rightarrow (n, t) \rightarrow y$
with $L = L_{PDE} + L_{FC} + L_{OC}$

$\Rightarrow$ SB2 $\rightarrow$ plot $\Rightarrow u(x, t)$
 $at \Rightarrow t = 0, 0.25, 0.5, 0.75, 1$

A SB3 $\rightarrow$ Calculate suitable global errors against benchmark / ref soln

$\Rightarrow$ SB4 $\rightarrow$ Uniform observe diff. $\rightarrow$ RAR-4 region
 $\hookrightarrow$ RAD $\rightarrow$ RAR-D

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5.1

Poisson $\rightarrow n \rightarrow u$

Burgers $\rightarrow (n, t) \rightarrow u$

2D Nav $\rightarrow (n, y) \rightarrow (u, v, p)$

3D Nav $\rightarrow (n, y, z) \rightarrow (u, v, w, p)$

HT-KL-W $\rightarrow (n, y, z) \rightarrow (u, v, w, p, k_T, k_v, \omega)$

2nd den

time + non linear conv.

multiple coupled fields + pressure

3D external flow

additional coupled transport eqn.

INACA 0021 $\rightarrow$ actual aerodynamic geometry & quantities

★ L_∞

$\Rightarrow$ let $u_n \rightarrow$ exact sol., $e(n) \rightarrow$ PINN gap

err. position $\Rightarrow e(n) = u_e(n) - u(n)$

$L_\infty \rightarrow$ what's largest absolute error anywhere in the point/domain evaluated.

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$$L_\infty = \max(|u_{0,i} - u_i|)$$

$$\textcircled{p} n_1 \rightarrow 0.001 \dots n_3 = 0.005 \dots$$

$$n_n \rightarrow 0.008$$

$$L_\infty = 0.008 \rightarrow \text{won't error}$$

$$L_2^{rel} = \frac{\|u_0 - u_2\|_2}{\|u\|_2}$$

global error

$L_2 \rightarrow$ How accu. it soln overall

$L_\infty \rightarrow$ How bad is worst pt.

in burgers

$\rightarrow$ steep region $\Rightarrow$ so uueny error might be decent but error might be large - around local region.

★ Burgers section from paper

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial n} = \nu \frac{\partial^2 u}{\partial n^2}, \quad n \in [-1, 1], \quad t \in [0, 1]$$

$$u(n, 0) = -\sin(\pi n)$$

$$u(-1, t) = u(1, t) = 0$$

$u \rightarrow$ flow velocity
 $\nu \rightarrow$ viscosity
 $\nu = 0.01 \frac{1}{\pi}$

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Res $\approx$ 1000 to 10,000

15000 iterations

with Adam opti. then 15000 iters. of L-BFGS

→ Manually method conv. fastest & results to lowest L^2 val. error among all uniform sampli. methods

→ Resampling using Random-R with 2000 res. points.

→ Random R outperforms Random method

When

Non-uniform → 15000 Adam then 10000 iters of L-BFGS adaptive methods

→ RAD → 2000 res. pts → resample every 2000 iterations → RAD gradient and Over random R
 (1000 Adam + 1000 L-BFGS) → RAD → $k=6$ on $C=0$
 L^2 barely dec. during training process

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(ie) both extreme cases show worse performance

for $k=1, k=1 \rightarrow L^2$ dec. rapidly

RAD is also ~~off~~. when choosing a set of k & C in moderate range

→ RAR-D → 1000 res. pts. in pre-trained process & added 1000 res. pts. for 2000 iterations (1000 Adam, 1000 L-BFGS)

until total no. of res. pts. reaches 2000

Optimal values for k & $C \Rightarrow 2 \& 0$

When using 2000 res. pts. both RAD & RAR-D have adv. over uniform sampling methods.

① When no. res. pts is fixed at 2000 all uniform sampling methods fail to capture to soln well.

Random-R performs best among uniform methods (1.69 ± 1.67)

Proposed RAD & RAR-D methods can achieve L^2 val error two orders of magnitude lower than that (0.027 .)

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⑧ $U_{nn} \rightarrow$ run lines (no)

$$\hookrightarrow \frac{d^2}{dx^2} (u_1 + u_2) = \frac{d^2 u_1}{dx^2} + \frac{d^2 u_2}{dx^2}$$

$\hookrightarrow$ linearity prop.

$x \rightarrow$ spatial position of

$u \rightarrow$ phys. qty. we are solving for

for burgers $\rightarrow u$ can be interpreted as a scalar flow velocity

u depends on x & t

⑨ at a specific x position what is the vel.

$u_{nn} \rightarrow$ spatial slope of velocity field

$$\left[\frac{d^2 u}{dx^2} \right] \rightarrow \text{second spatial derivative} \rightarrow \text{spatial curvature}$$

$\lambda \rightarrow$ operational hyperparameter

$\nu \rightarrow$ det. how strongly the diffusion/viscous effects act

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$$u_t + u u_n = \nu u_{nn}$$

$u u_n \rightarrow$ non-linear transport / advection

$\nu u_{nn} =$ Viscous smoothing / diffusion

Poisson $\Rightarrow u_{nn} = -\pi^2 \sin(\pi x) \rightarrow$ Steady boundary-value problem

Burgers $\Rightarrow u_t + u u_n = \nu u_{nn} \rightarrow$ time dep. non-linear PDE

Poisson $\Rightarrow u(x) = \sin(\pi x) \rightarrow$ exact soln

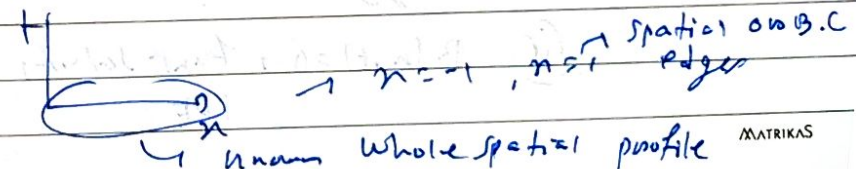
$$\therefore \frac{d^2}{dx^2} \sin(\pi x) = -\pi^2 \sin(\pi x)$$

Burgers $\Rightarrow u(x, 0) = -\sin(\pi x) \rightarrow$ I.C.
 $\nearrow$ initial vel. profile

$\hookrightarrow$ it only tells us what vel. profile is at $t=0$

$$u(x, 0) = -\sin(\pi x)$$

$\hookrightarrow$ at $t=0$, tell the kernel what vel. should be at every loc. x .



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Spatial → where the cond. is applied is $n = -1$

Dirichlet → what kind of qty. you prescribe

(a) $u(-1, t) = 0$

Neumann: $\frac{\partial u}{\partial n} = g \Rightarrow$ prescribe a derivative

Burgers PDE → spatial + Dirichlet

(b) Functⁿ $u(n, t)$ is req. to take the value 0 at spatial pt. $n = -1$ for all time t

Residual → how badly does the predicted Functⁿ violate the gov. PDE at this loc.

$R(n, t) = u_t + u u_n - \nu u_{nn}$

for Burgers

(c) $R(n, t) = 0 \rightarrow$ Functⁿ satisfies PDE

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(d) $R(n, t) = 0.5$ The pred. val. is 0.5 away from true val.

PINN → Network doesn't know actual soln →

"Whatever Functⁿ you produce must satisfy these physical eqns & these condit^{ns}"

(e) $u_t + u u_n - \nu u_{nn} = 0, u(n, 0) = \sin(n, \pi)$
 $u(-1, t) = u(1, t) = 0$

The network ini. produces a basically arbitrary Functⁿ, then they adjust weights so that its Functⁿ inc. satisfies these constraints

(f) $(n, t) \rightarrow u_0(n, t)$

i/p → randomly ini. network Functⁿ → some non-zero

then → does it satisfy PDE
 calc $\hat{R}(n, t)$

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when BC? $\Rightarrow$ calc. $u_0(-1, t)$

$\hookrightarrow u_0(1, t)$

when IC? $\Rightarrow$ calc. $u_0(x, 0) = [-\sin(\pi x)]$

then combine
those per. into
loss

$\Rightarrow$ I/p $\rightarrow$ train net
 $\rightarrow$ function that app.
solution to PDE $\neq$ ICLT
BC

so netw. learn soln from constraints.

Problem $u(x) = \sin(\pi x) \rightarrow$ analytic soln

$\Rightarrow$ Comp. $u_0(x)$ to $u(x)$

Burger $\rightarrow$ soln change with time

$\Rightarrow$ we need ref. soln from benchmark
or dataset.

$u_{PINN}(x, t)$ & $u_{ref}(x, t)$

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① $\|e\|_2 \rightarrow$ 2 norm

$e = [e_1, e_2, \dots, e_N]$

$$\|e\|_2 = \sqrt{e_1^2 + e_2^2 + \dots + e_N^2}$$

$$\textcircled{2} L_2 = \frac{\|u_0 - u_{exact}\|_2}{\|u_{exact}\|_2}$$

$\Rightarrow$ 2D $\rightarrow (x, t)$, flow problem is still 1D
spatially.

$$x \in [-1, 1]$$

$$t \in [0, 1]$$

③ 1D time-dependent Burgers eqn

$\textcircled{4} -\sin(\pi x)$ at $t=0 \rightarrow$ the problem is
defined this way
 $u(x, 0) = -\sin(\pi x)$

(ie) it's a benchmark

"at the initial we start the simulation, the
vel. distribution along the whole line
looks like this"

$-\sin(\pi x)$
initial condition

$x = -1 \Rightarrow u = 0$
 $x = -0.5 \Rightarrow u = 1$
 $x = 0 \Rightarrow u = 0$
 $x = 0.5 \Rightarrow u = -1$
 $x = 1 \Rightarrow u = 0$

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Weights/Biases

$$1 \text{ I/p} \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 1 \text{ O/p}$$

$$Z_1 = \underbrace{w_1}_{\text{weights}} \cdot x + \underbrace{b_1}_{\text{biases}}$$

then: $a_1 = \tanh(z_1)$

$$1^{\text{st}} \text{ layer} \Rightarrow 1 \text{ I/p} \rightarrow 32 \text{ nodes}$$

$$32 \text{ weights} + 32 \text{ biases} = 64$$

$$2^{\text{nd}} \text{ layer} \Rightarrow 32 \rightarrow 32 \Rightarrow 32 \times 32 + 32 = 1056$$

$$3^{\text{rd}} \text{ layer} \Rightarrow 1056 \quad 32 \times 32 + 32 = 1056$$

$$0 \text{ I/p layer} \Rightarrow 32 \rightarrow 1 \Rightarrow 32 + 1 = 33$$

$$= 2201$$

Optimiser step 1)
change from
2201 weights

Bugan high res num. ref. soln

$$\text{Bugan} \Rightarrow 2 \rightarrow 32 \rightarrow 32 \rightarrow 32 \rightarrow 1$$

$$\hookrightarrow 2241$$

$$\text{Let } x = 0.3 \text{ \& } t = 0.7$$

$$\Rightarrow [0.3, 0.7] \rightarrow \text{Feetly}$$

$$U_0(0.3, 0.7) \approx 0.42$$

o/p of current w & b

$$\text{then } R = U_t + U_{t+1} - U_{t-1}$$

if $R \uparrow$ Adam changes weights.

For ref soln $\Rightarrow$ using conv. numerical solvers

PDE $\rightarrow$ High res num. solvers $\rightarrow U_{\text{ref}}(x, t)$

PINN $\rightarrow U_0(x, t)$

for comp. $[U_0 \text{ vs } U_{\text{ref}}]$

lets not download random ref & call it ground truth.

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① IC of $u(m, 0) = -\sin(m\pi)$

PINN model $\Rightarrow u_a(n, t)$

$u_{ref}(n, t) \Rightarrow$ trusted num approx. of actual
Burgers soln. used after
during evalua, not as to PINN
training lab.

② PDF + IC + BC

↓
PINN training

↓

$u_a(n, t)$

then

Evaluate

$u_\theta(n, t)$

$u_{ref}(n, t)$

error. why?

⊛ How Ref soln will be generated

L_2 can be used as our ref. soln

∵ Under train setup it was $L_2 = 0.2$

⊛ ⇒ Repo... by Raissi on sampling papers

⊛

dt.